

(20 / marks)

Part A: Multiple choice

-Be sure that you have 10 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1) The solution of the differential equation $y' - 2 = 6x + y$ is

- a)
- $y = -6x + ce^3 - 9$
- b)
- $y = x + ce^x$
- c)
- $y = ce^{2x} + 3x + 1$

d) $y = -6x + ce^x - 8$

Q2) The solution of the IVP $y'' + 3y' - 4y = 0$, $y(0) = 1$, $y'(0) = 2$ is

a) $y = 6e^{2x} - \frac{1}{5}e^{-4x}$

b) $y = \frac{6}{5}e^{-x} - \frac{1}{5}e^{4x}$

c) $y = \frac{6}{5}e^x + \frac{1}{5}e^{-4x}$

d) $y = \frac{6}{5}e^x - \frac{1}{5}e^{-4x}$

Q3) The solution of the differential equation $y'' + 4y' + 4y = 0$ is

a) $y = c_1e^{4x} + c_2xe^{2x} + c_3$

b) $y = c_1e^{-2x} + c_2xe^{-2x}$

c) $y = c_1 + c_2e^{-2x} + c_3xe^{-2x}$

d) $y = c_1 + c_2e^{2x} + c_3xe^{2x}$

Q4) If the roots of characteristic equation of a homogeneous linear differential equation with constant coefficients are $\lambda = -2, \pm 2i$, the differential equation is

a) $y'' + 2y' + 8y = 0$

b) $y'' + 2y' + 4y + 8 = 0$

c) $y'' + 2y' + 4y' + 8y = 0$

d) $y'' - 2y'' - 4y' - 8y = 0$

Q5) Find the second solution for $x^2y'' - 2xy' - 4y = 0$ such that $y = x^4$

a) $y_2 = \frac{-2}{5x}$

b) $y_2 = -\frac{1}{5x}$

c) $y_2 = \frac{5}{x}$

d) $y_2 = \frac{x}{5}$

Q6) Let $y = x^m$ solutions of the differential equation $x^3y'' - 2xy' - 10y = 0$ the values of m

a) 4, -1

b) -2, 5

c) -2, 4

d) 3, -2

Q7) The solution of the differential equation $(ye^y - 2y^3)dx + (xe^y - 6xy^2 - 2y)dy = 0$ is

a) $e^y - 2xy^3 - y^2 = c$

b) $e^y + 2xy^3 + y^2 = c$

c) $e^y - 2xy - 3y^2 = c$

d) $e^y - 2xy^3 - y^2 = c$

Q8) The solution of the differential equation $\frac{1}{x}y' - \frac{2}{x^2}y = x \sin x$ is

a) $y = -x^2 \cos x + cx^2$

b) $y = -\cos x + cx^2$

c) $y = -x \cos x + cx^2$

d) $y = x^2 \cos x - cx$

Q9) The solution of the differential equation $y' - xy = 4e^{\frac{1}{4}x^2} \sqrt{y}$ is

a) $y = -e^{\frac{x^2}{4}} + ce^{\frac{x^2}{4}}$

b) $y = 2xe^{\frac{x^2}{4}} + ce^{\frac{x^2}{4}}$

c) $y = \left(2xe^{\frac{x^2}{4}} + ce^{\frac{x^2}{4}} \right)^2$

d) $y = \left(2e^{\frac{x^2}{4}} + ce^{\frac{x^2}{4}} \right)^2$

Q10) The value of a make $f(x) = \ln x$ a solution to $(3xy')^2 + a = 3$ is

a) 1

b) -4

c) -1

d) -6

Part C: Written Questions

(10 / marks)

Q11: The general solution of differential equation is $y = x^4(c_1 \cos(4 \ln x) + c_2 \sin(4 \ln x))$ find the differential equation.Q12: Solve $x^2y'' + 2xy' - 20y = 0$

Best Wishes /

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
d	d	c	No	b	b	a	a	c	d

11) $y = c_1 x^2 \cos(4 \ln x) + c_2 x^2 \sin(4 \ln x)$

$$m = \alpha + \beta i$$

$$\alpha = 2$$

$$\beta = 4$$

$$m = 2 \pm 4i$$

$$(m-2)^2 \pm 4i)^2$$

$$m^2 - 4m + 20$$

$$m^2 - 4m + 4 = 16$$

$$m^2 - 4m + 20 = 0$$

$$m^2 - 4m + 3m - 3m + 20 = 0$$

$$m^2 - m - 3m + 20 = 0$$

$$x^2 y'' - 3x y' + 20y = 0$$

12)

Sol

$$x^2 y'' + 2x y' - 20y = 0$$

$$m(m-1) + 2m - 20 = 0$$

$$m^2 - m + 2m - 20 = 0$$

$$m^2 + m - 20 = 0$$

$$m = \frac{-1 \pm \sqrt{1 - 4(-20)}}{2}$$

$$= \frac{-1}{2} \pm \frac{\sqrt{81}}{2}$$

$$m = \frac{-1}{2} \pm \frac{9}{2} \Rightarrow$$

$$m = \frac{-1}{2} + \frac{9}{2}, m = \frac{-1}{2} - \frac{9}{2}$$

$$m = 4, m = -5$$

$$\Rightarrow y = c_1 x^4 + c_2 x^{-5}$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$